Requirements:

Test tube, Iodine solution, Extract of sample foodstuff (potato, rice, wheat or maize grains).

Procedure:

1. Take 1 ml of sample solution in a test tube.
2. Add 1 ml of Iodine (Potassium iodide).
3. Appearance of blue-black colour.

Table:

Procedure	Observation	Inference
1ml of sample solution + 1ml of Iodine solution	Appearance of deep blue-black colour	Starch is present

Exercise: 30

Test for protein – Biuret test

Aim :

To detect the presence of proteins.

Basic Principle:

1. Proteins are polymer of amino acids. (Polypeptide).
2. Amino group of one amino acid binds with carboxylic group of another amino acid to form peptide bond. (NH-CO linkage)
3. In alkaline medium $CuSO_4$ reacts with peptide bond and gives a purple colour .
4. All proteins do not contain the same amino acids, and hence they do not respond to all colour reactions. (Biuret test is for peptide bond in the molecule of a protein, xanthoproteic test is specific for protein containing aromatic amino acids).

Requirements:

Test tube, NaoH, $CuSO_4$ solution, milk/albumin of egg / gram seed extract.

Procedure:

1. Take 2 ml of sample solution.
2. Add 1 ml of sodium hydroxide solution.
3. Add 1 or 2 drops of 1% copper (II) sulphate and mix it well.
4. Appearance of Purple colour (Increase with increase in concentration)

Table:

Procedure	Observation	Inference
2 ml of sample solution + 1 ml of Sodium hydroxide + 1 or 2 drops of 1% Copper (II) sulphate and mix it well.	Appearance of Purple colour	Protein is present

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